Status	Finished
Started	Tuesday, 20 October 2026, 10:08 PM
Completed	Tuesday, 20 October 2026, 10:15 PM
Duration	6 mins 14 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all

b. When the item is the last element in the array

c. When the item is not the array at all

d. When the item is somewhere in the middle of the array ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list

b. It works on unsorted lists

c. It compares elements sequentially

d. It can be applied only if the list is sorted ✓

Question **3**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)

b. Linear search ✓

c. Hash search

d. Binary search

Question **4**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the middle of the list
b. In the left sublist
c. At the beginning of the list
d. In the right sublist ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list is sorted ✓
c. When the list contains duplicate elements
d. When the list is unsorted

Question **6**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. n ✓
c. 1
d. n/2

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By comparing each element in the list sequentially ✓
b. By using a hash function
c. By dividing the list into halves
d. By sorting the list first

Question **8**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Sorting elements in a list
- c. Deleting elements from a list
- d. Inserting elements into a list

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Linear search ✓
- b. Quick search
- c. Merge search
- d. Bubble search

Question **10**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(log n)
- b. O(1) ✓
- c. O👎
- d. O(n log n)

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It does not work with negative numbers
- b. It requires the list to be sorted ✓
- c. It is slower than linear search for small lists
- d. It can only be applied to large lists

Question **12**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 10 ✓
- b. 1000
- c. 100
- d. 500

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the entire list
- b. At the beginning of the list
- c. In the right sublist
- d. In the left sublist ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Finding
- b. Searching ✓
- c. Discovering
- d. Mining

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. There must be a mechanism to delete and/or insert elements in the list ✓
- b. The list must be sorted
- c. There should be direct access to the middle element in any sublist
- d. Number values should only be present

Number values should only be present